FACULTY OF ENGINEERING
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EE568 - Selected Topics on Electrical Machines

2025-2026 Spring Semester

Project - 1

Report

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Introduction

This is the report of the first project for course EE 568 Selected Topics on Electrical Machines. Project consist of 6 different questions. Purpose of this project is gaining analytical understanding on variable reluctance machines and learn how to simulate them using FEA software.

In the first question we asked for analytical solution of a singly excited variable reluctance machine with numerical values.

In Q2 we asked to do a linear simulation of this machine. For simulation purposes I use Ansys Maxwell with Metu Student License.

In Q3 we asked to do a nonlinear simulation of this machine. Again, for simulation p purposes I use Ansys Maxwell with Metu Student License.

In Q4 we asked to propose a control method to enable continuous rotation. I explain my solution with a graph which is drawn with python.

In Q5 we asked to animate the rotation of this machine. Ansys Maxwell is a powerful tool for animation.

In Q6 we asked to repeat the simulation with 3D nonlinear. Since I used Ansys Maxwel from start all my simulations are 3D.

All the files related to this project report are available at:

https://github.com/utkutetikce/EE568Project1

Q1)

A)

In order to calculate the maximum and the minimum reluctance (and inductance) values as the rotor makes a full rotation analytically we need to make some assumptions.

- Infinite core permeability
- All leakage flux neglected
- All fringing flux neglected

The equations for maximum and minimum reluctance are:

$$R_{min} = \frac{2.\, l_{airgap(min)}}{\mu_0 .\, A_{eff.\ area}}$$

$$R_{max} = \frac{2.\, l_{airgap(max)}}{\mu_0 .\, A_{eff.\ area}}$$

Since all fringing flux neglected $A_{eff.\ area}$ are same as both equation. We can calculate the $A_{eff.\ area}$ by using simple trigonometric equations. Since our core has hole shaped as a circle with radius of 12.5 mm and our core have a width of 15mm and depth of 25mm we can derive same basic triangles and find radius which give us the arc length:

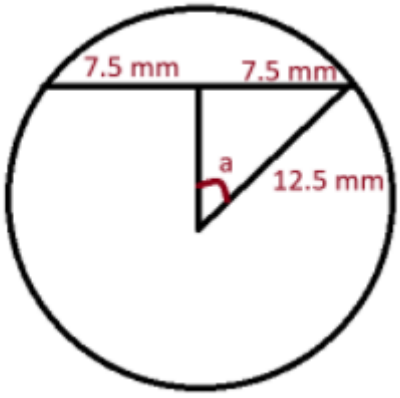


Figure 1 Circle Parameters of Core Hole

$$a = \sin^{-1}\left(\frac{7.5}{12.5}\right) = 36{,}86° \ , 2a = 73{,}72°$$

Then convert this angle to radian to calculate arclength:

$$A_{eff.\ area} = 25mm\,.\left(\frac{\pi}{180}\right).\,12{,}5mm\,.\,73{,}72° = 402{,}1\ mm^2$$

Since our $l_{airgap(min)} = 0.05mm$ and $l_{airgap(max)} = 2.5mm$ we can calculate the reluctance values:

$$R_{min} = \frac{2.\, l_{airgap(min)}}{\mu_0 . A_{eff.\ area}} = \frac{2.\,(0.05.\,10^{-3})}{(4\pi.\,10^{-7}).\,402{,}1.\,10^{-6}} = 1979046{,}8\ H^{-1}$$

$$R_{max} = \frac{2.\, l_{airgap(max)}}{\mu_0 . A_{eff.\ area}} = \frac{2.\,((2.5).10^{-3})}{(4\pi.\,10^{-7}).\,402{,}1.\,10^{-6}} = 9895234\ H^{-1}$$

Since we calculated reluctance values and we know the N = 300, we can calculate inductance values as well:

$$L_{max} = \frac{N^2}{R_{min}} = \frac{300^2}{1979046{,}8\ H^{-1}} = 0{,}045\ H$$

$$L_{min} = \frac{N^2}{R_{max}} = \frac{300^2}{9895234\ H^{-1}} = 0{,}0091\ H$$

Since we calculated the L_{max} and L_{min} values we can plot the inductance values as a function of rotor angle. First we calculate these values:

$$L_1 = \frac{L_{max} + L_{min}}{2} = 0.027$$

$$L_2 = \frac{L_{max} - L_{min}}{2} = 0.018$$

Then we can draw the inductance plot with:

$$L(\theta) = L_1 + L_2 \cos(\theta)$$

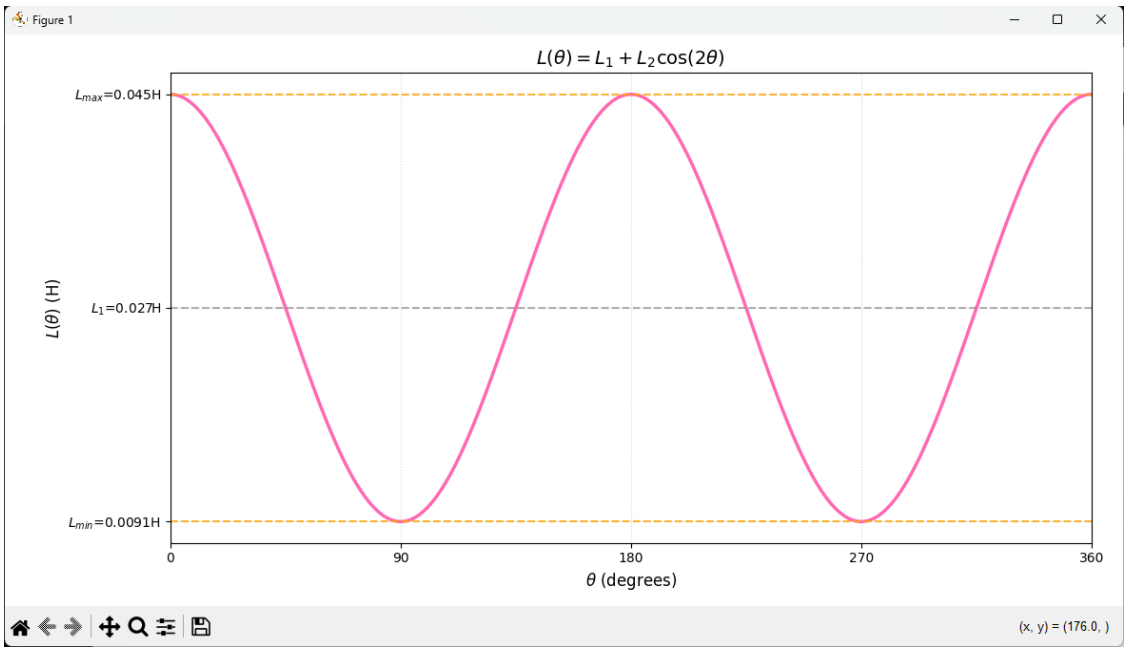


Figure 2 Inductance Over Rotor Angle(Python source code available on GitHub)

B)

The torque generated in the system can be described with this equation:

$$T = \frac{1}{2}.i^2.\frac{dL(\theta)}{d\theta}$$

We can write torque as a function of rotor angle with this equation:

$$T(\theta) = -i^2 L_2 \cos(2\theta)$$

Since we derived the inductance plot we can derive torque plot using python:

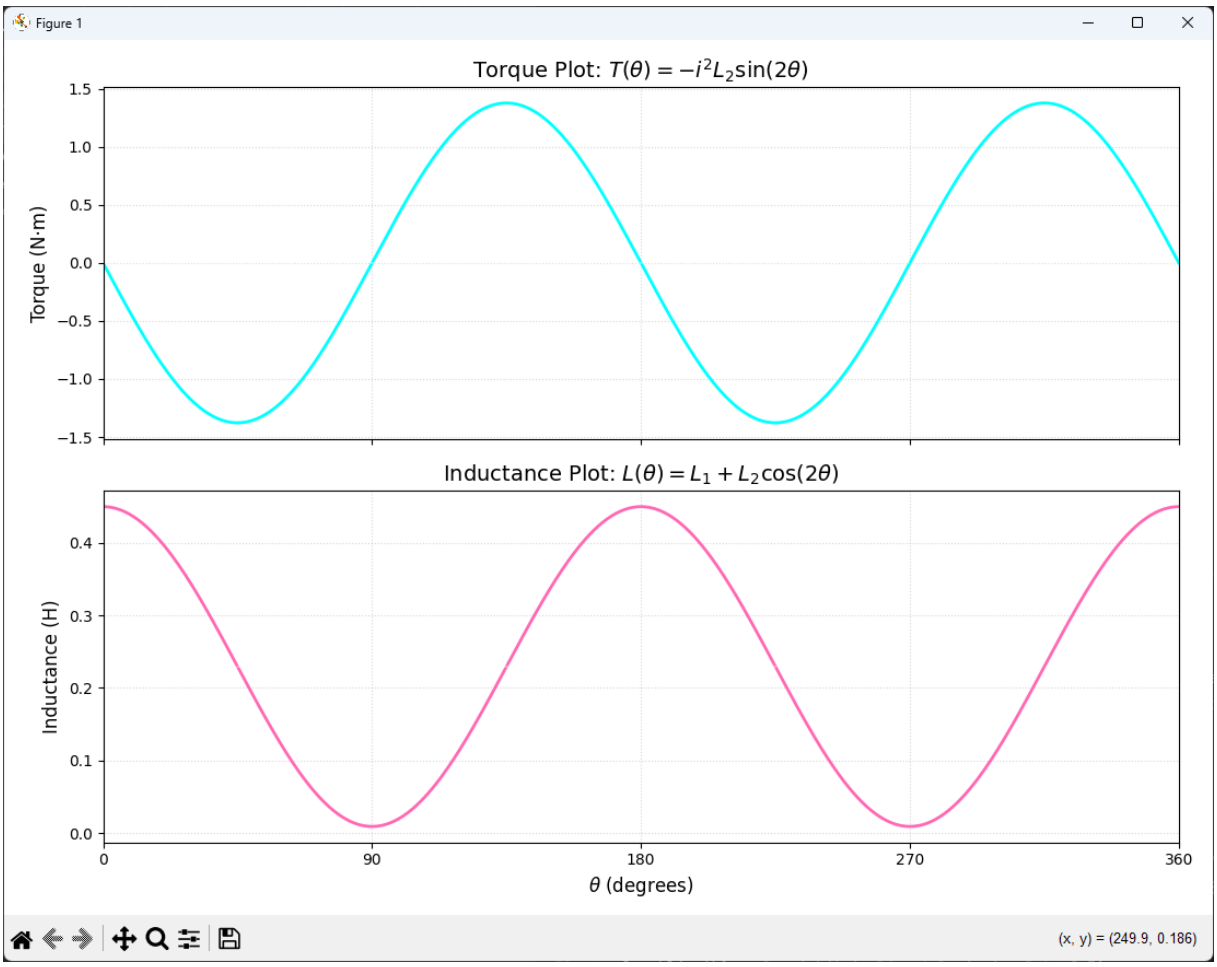


Figure 3 Torque and Inductance Graphs(Python source code available on GitHub)

By looking at these graphs we can see the relationship between torque and inductance. For example when we set the rotor angle at 80° we can see there is a negative torque applying to the rotor which will move the rotor towards 0° position at the path towards 0° the inductance of the system will be reduce, which will reduce the torque generated in the system and that exactly 0° there will be no torque generated in the system. The system will be at equilibrium point.

C)

This electric machine is singly excited variable reluctance machine. Since the rotor has a one position, no matter the current direction rotor only rotates until rotor position where the torque is 0. Rotor will pass this rotor angle but it will come back to the same position after few oscillations. Applying current with pulses which will apply torque to the rotor when it rotates 90° to 180° and 270° to 360°. This way we can obtain an continuous rotation.

Q2)

For modeling the variable reluctance machine for this project, I choose Ansys Maxwell as a FEA program. After downloading Ansys Maxwell from Metu Software Center, I began my simulation. First I draw 4 box shaped objects then using Boolean Unite function to create a core which is look like:

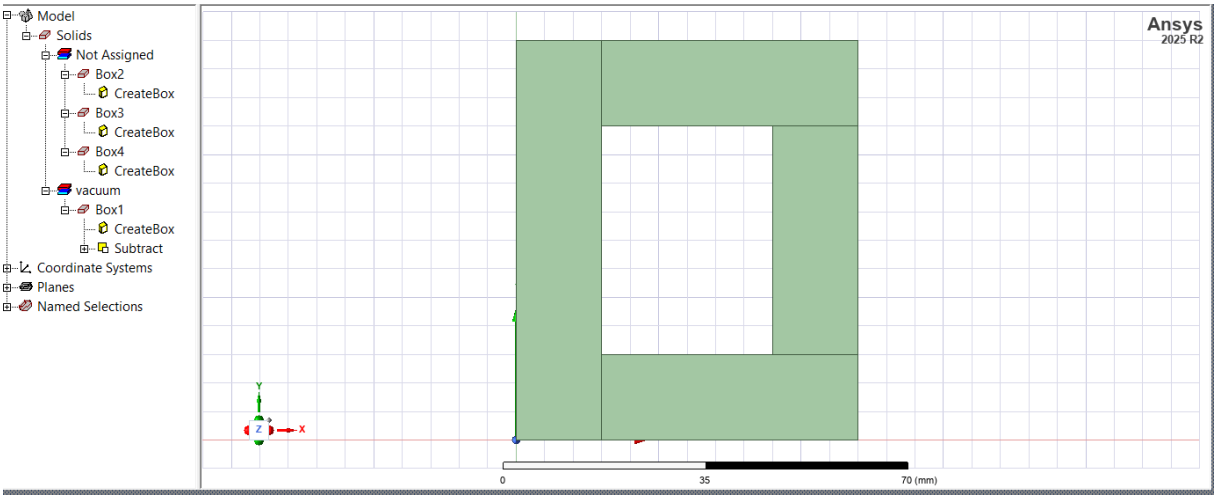


Figure 4 Stator Structrue View Before Unite

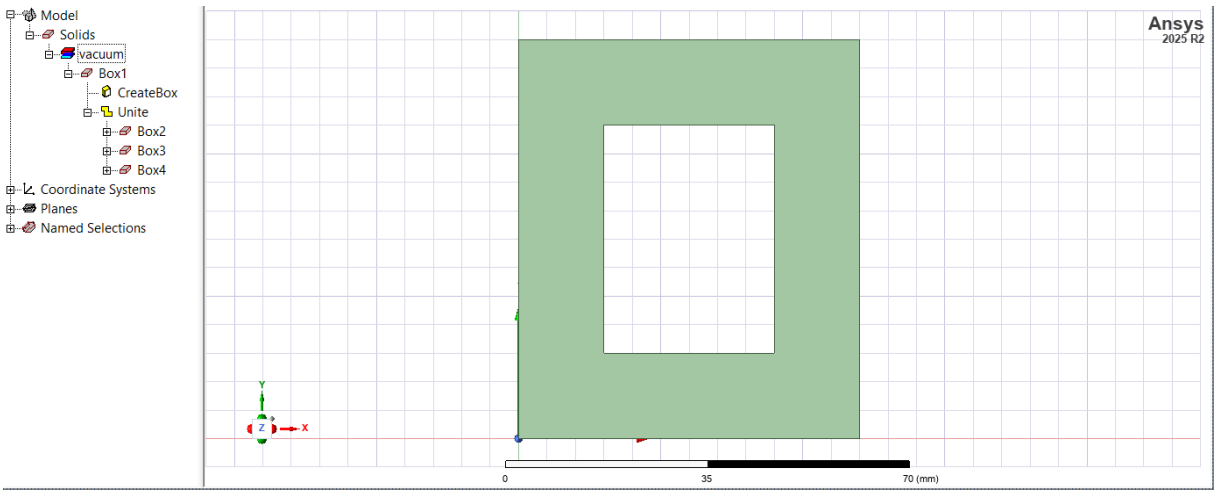


Figure 5 Stator Structure View After Unite

After creating the basic core layout I create a cylinder and use Boolean Subtract function to create the final Stator form.

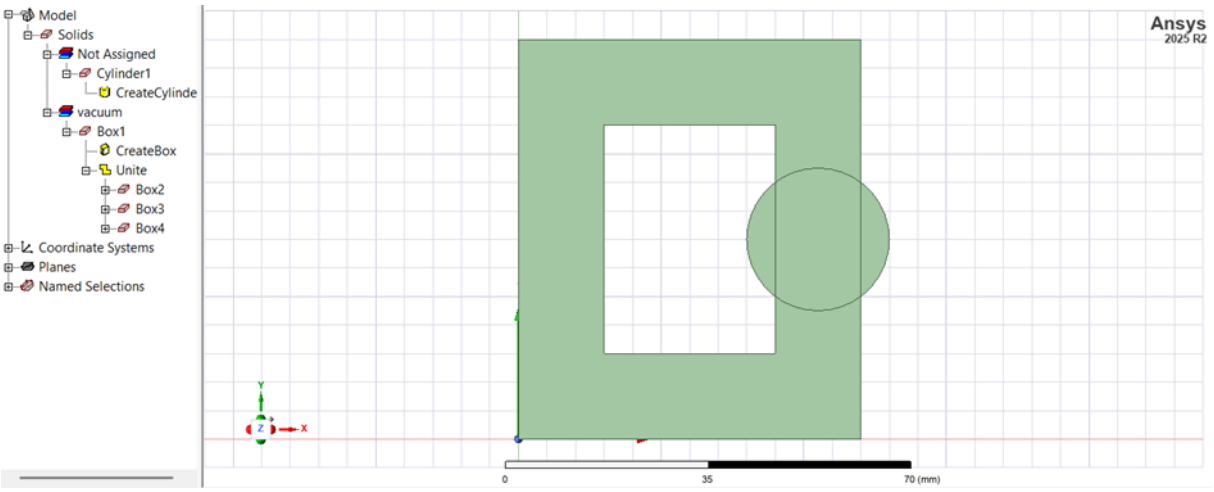


Figure 6 Stator Structure View Before Subtract

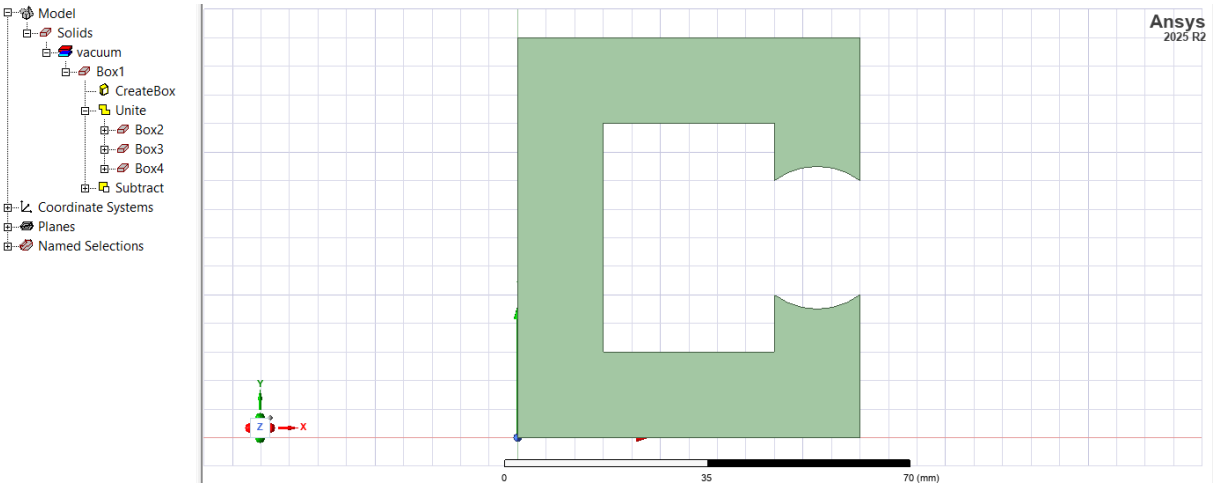


Figure 7 Stator Structure View After Subtract

After I created the stator, I use the same steps for rotor. Using 2 cylinders and 4 rectangles and applying unite, subtract and intersect functions this time I get the wanted rotor geometry.

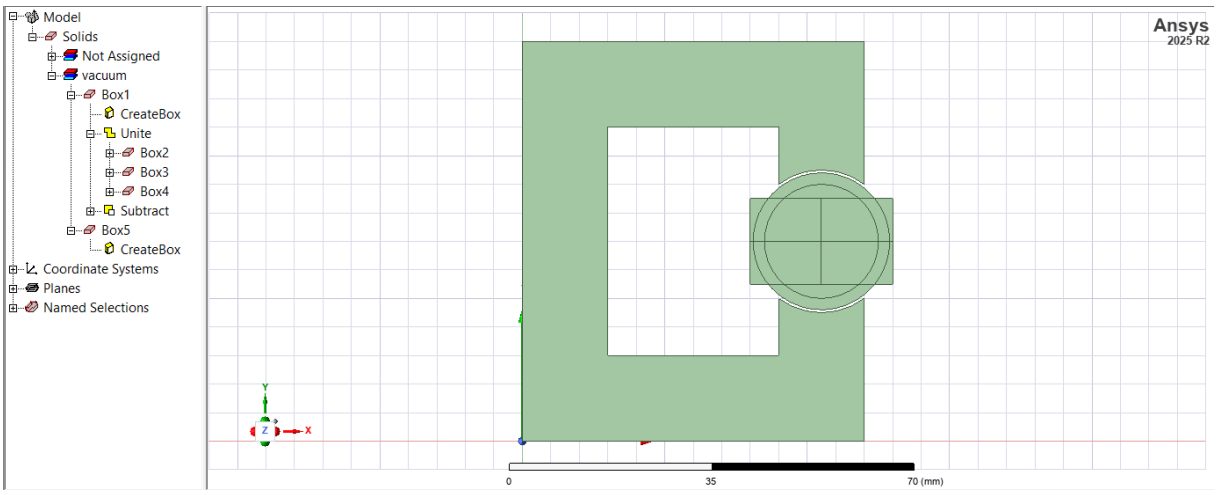


Figure 8 Rotor Structure View Before Unite, Subtract and Intersect

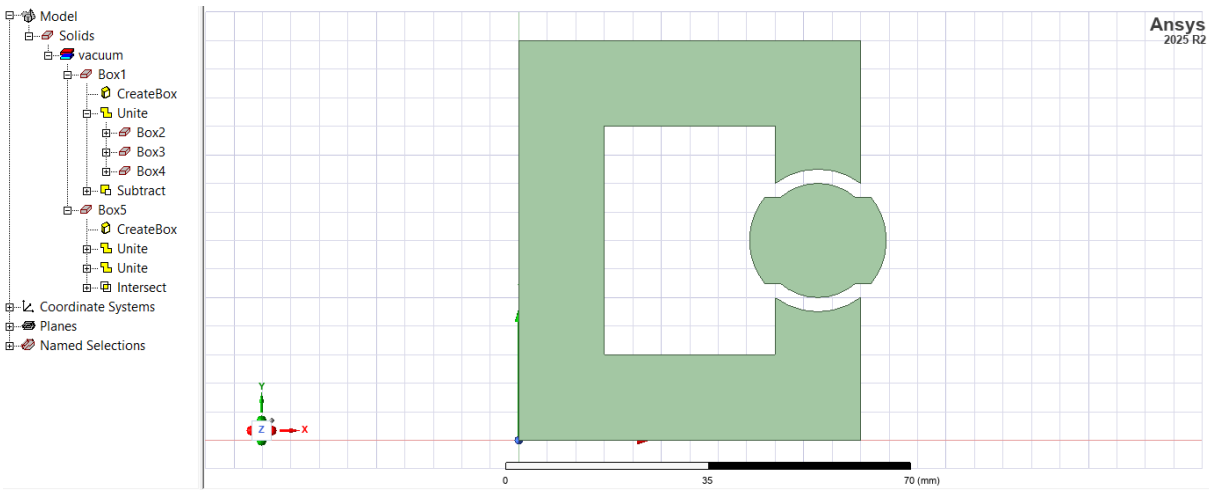


Figure 9 Rotor Structure View After Unite, Subtract and Intersect

Now I have created desired stator and rotor geometry. Note that since I am using Ansys Mawell all my designs are 3D.

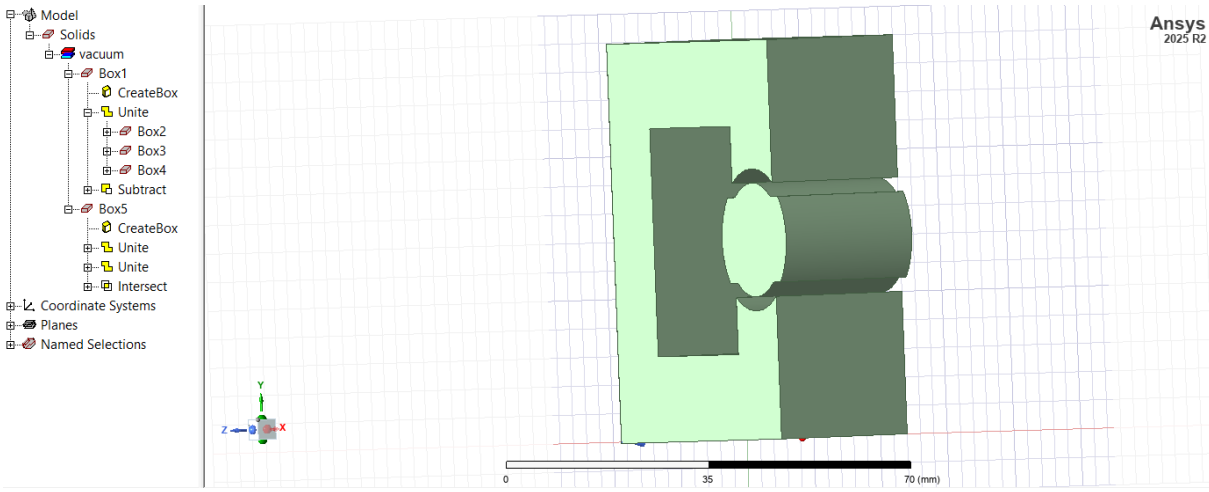


Figure 10 3D View of Stator and Rotor

Next step is Adding Coils for our simulation. Since it is been said that "You can approximate the coils by solid copper regions" in project details, I will model the coils as a one single solid cooper region. Also I changed the colors of stator, rotor and winding in different colors so it is easier to recognize them.

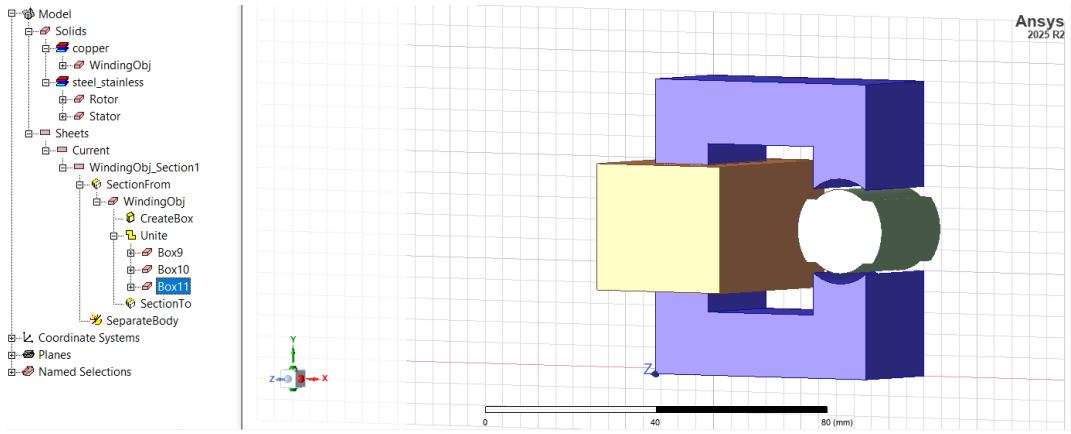
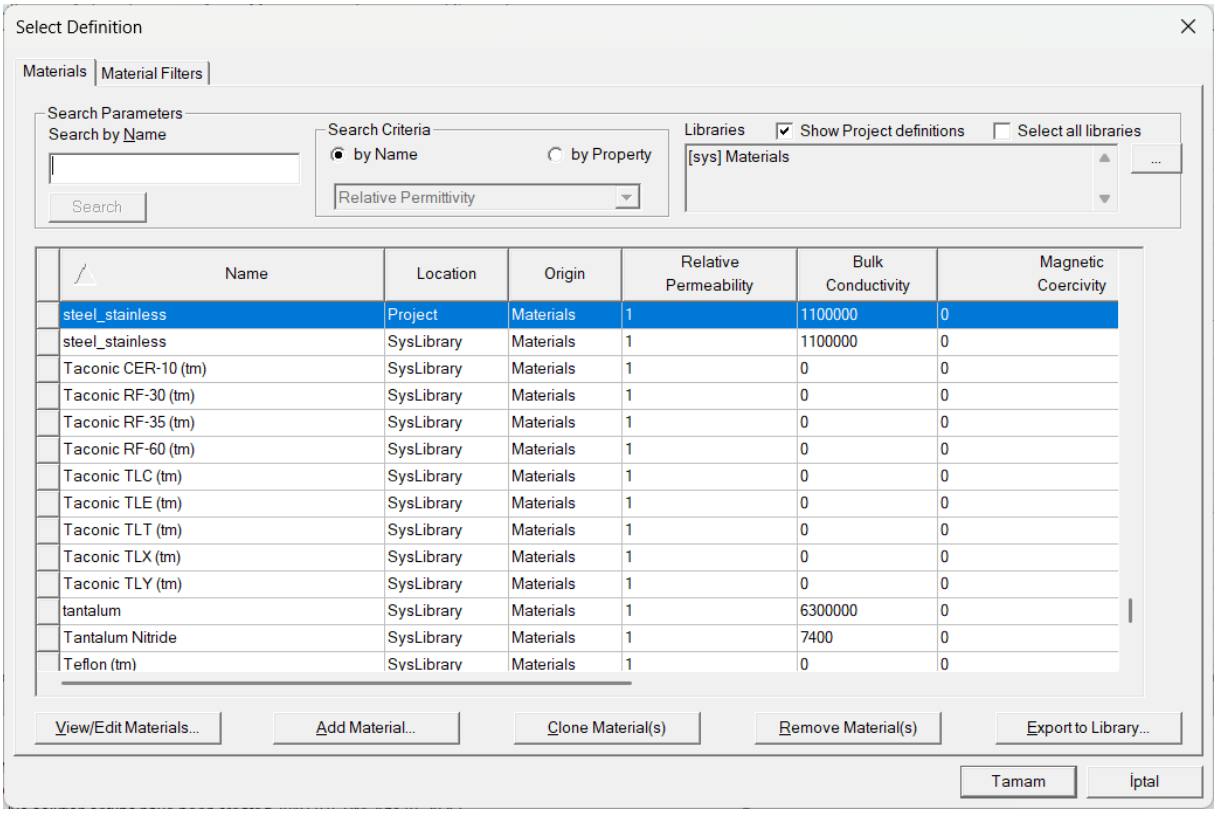


Figure 11 3D View of Stator, Rotor and Winding

In Q2 FEA modelling will be done with linear materials.



Name	Location	Origin	Relative Permeability	Bulk Conductivity	Magnetic Coercivity
steel_stainless	Project	Materials	1	1100000	0
steel_stainless	SysLibrary	Materials	1	1100000	0
Taconic CER-10 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-30 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-35 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-60 (tm)	SysLibrary	Materials	1	0	0
Taconic TLC (tm)	SysLibrary	Materials	1	0	0
Taconic TLE (tm)	SysLibrary	Materials	1	0	0
Taconic TLT (tm)	SysLibrary	Materials	1	0	0
Taconic TLX (tm)	SysLibrary	Materials	1	0	0
Taconic TLY (tm)	SysLibrary	Materials	1	0	0
tantalum	SysLibrary	Materials	1	6300000	0
Tantalum Nitride	SysLibrary	Materials	1	7400	0
Teflon (tm)	SysLibrary	Materials	1	0	0

Figure 12 Linear Material Selection

After selecting Linear materials for stator, rotor and winding our next step is to create a Excitation which is the representation of the current that we gave to the coils in order to create flux on the magnetic circuit. To do that first I chose the winding object (Which is a

single rectangular object in this project) and create an isolating layer on the surface of this object. Than create a surface inside this object which acts as a current flow enclosure. Then adjust the current value which is passing from this enclosure. Note that since I approximate the coils as a single solid cooper object, current value I entered to this parameter is 750 A (300 x 2.5 A = 750 A).

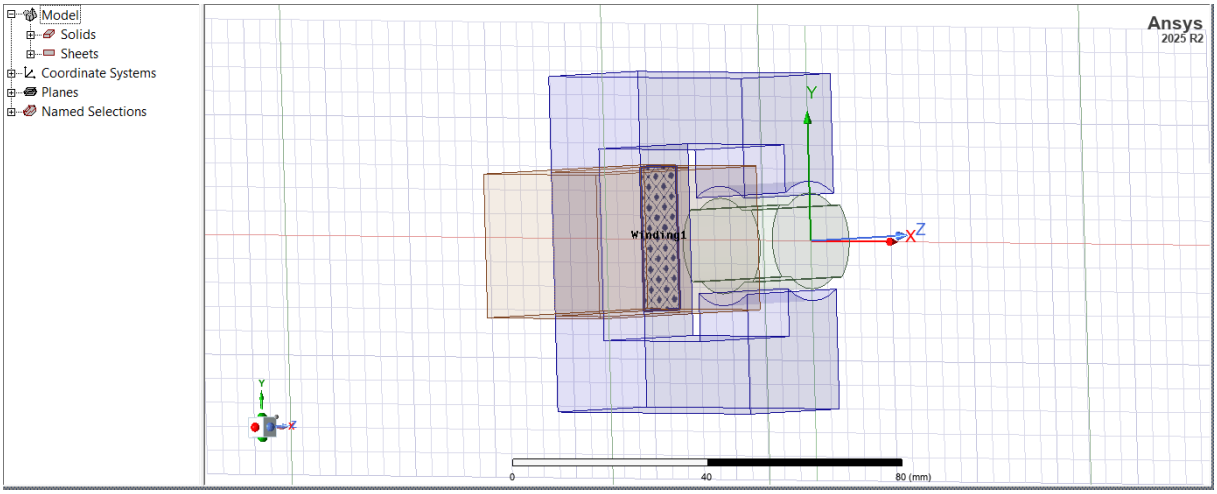


Figure 13 Coil Current Surface View

After that I change the origin point of coordinate system on simulation in order to define the rotor rotation. After that I define an rotate function under the rotor and create a variable named rotorAngle which is the angle of rotor relative to the origin.

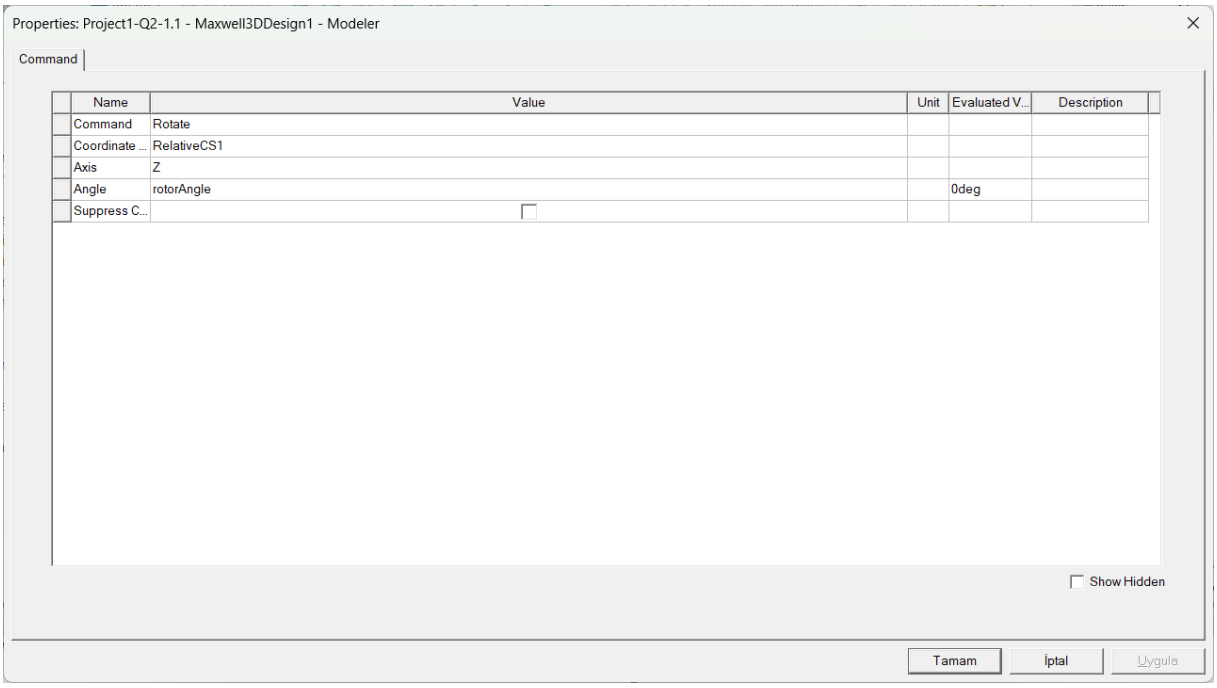


Figure 14 rotorAngle Variable

After creating rotorAngle variable I create a new ParametricSetup.

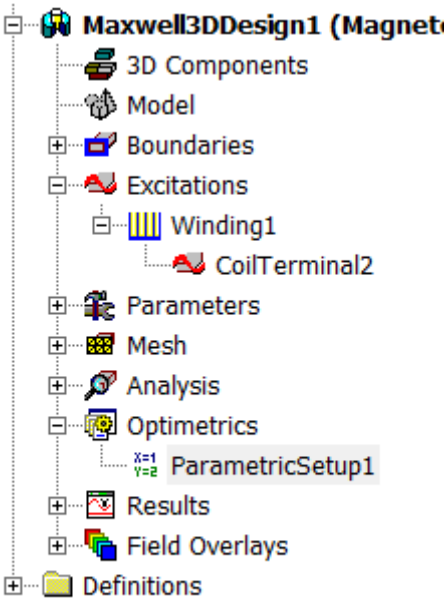


Figure 15 ParametricSetup on Project Manager

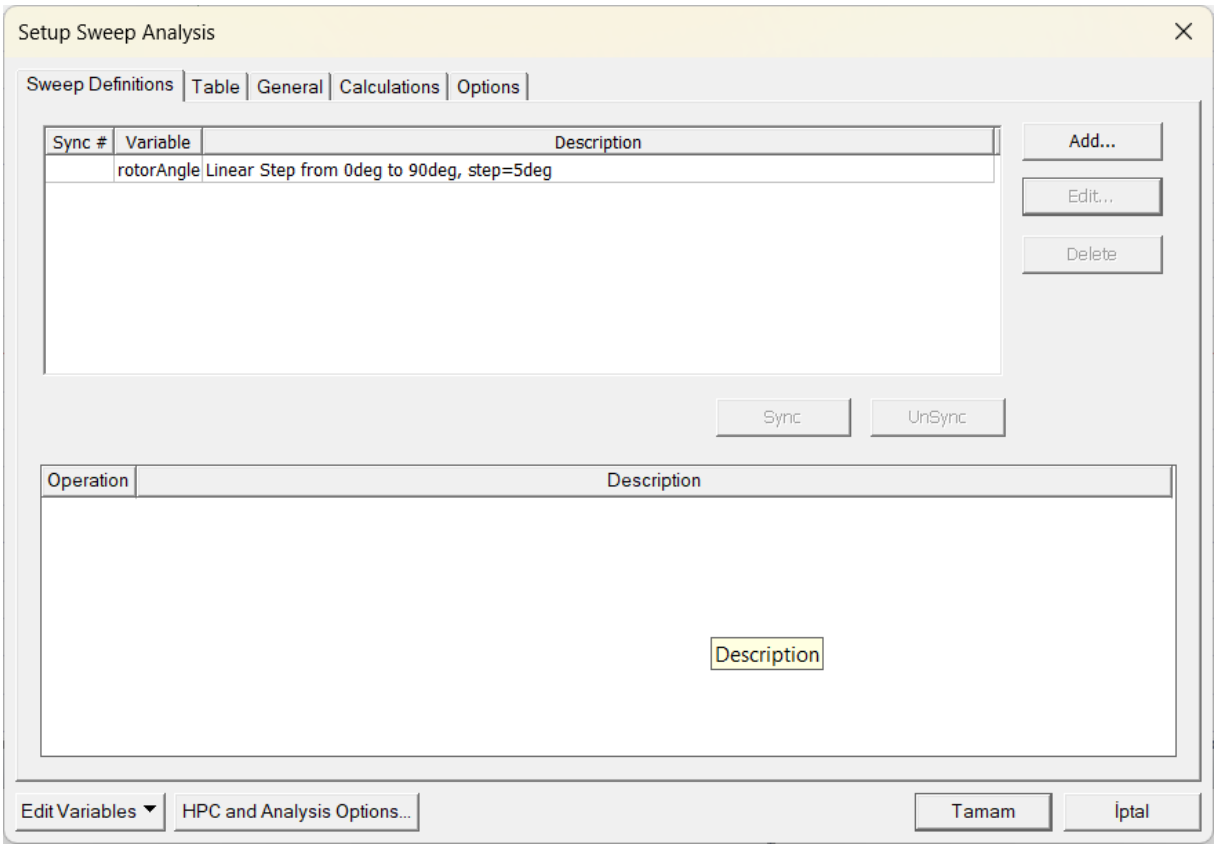


Figure 16 rotorAngle Sweep parameters

A)

By looking at the flux density vectors for three positions we can clearly see that flux density vectors change depending on the rotor position(specially on rotor). When rotor angle is 0° the magnetic circuit faces maximum reluctance due to the widest possible air gap. Which results a fringe effect on flux density vectors. Flux vectors spread out (fringe) widely, resulting in a less concentrated field and a slight drop in peak flux density. When rotor angle is 90° the magnetic circuit faces minimum reluctance due to the shortest possible air gap. The minimized air gap creates a path of least resistance, allowing the flux to flow in a straight, highly concentrated manner that yields the system's highest maximum flux density. When rotor angle is 45° the air gap becomes asymmetrical, forcing the flux density vectors to bend aggressively and distort as they seek the nearest highly permeable steel.

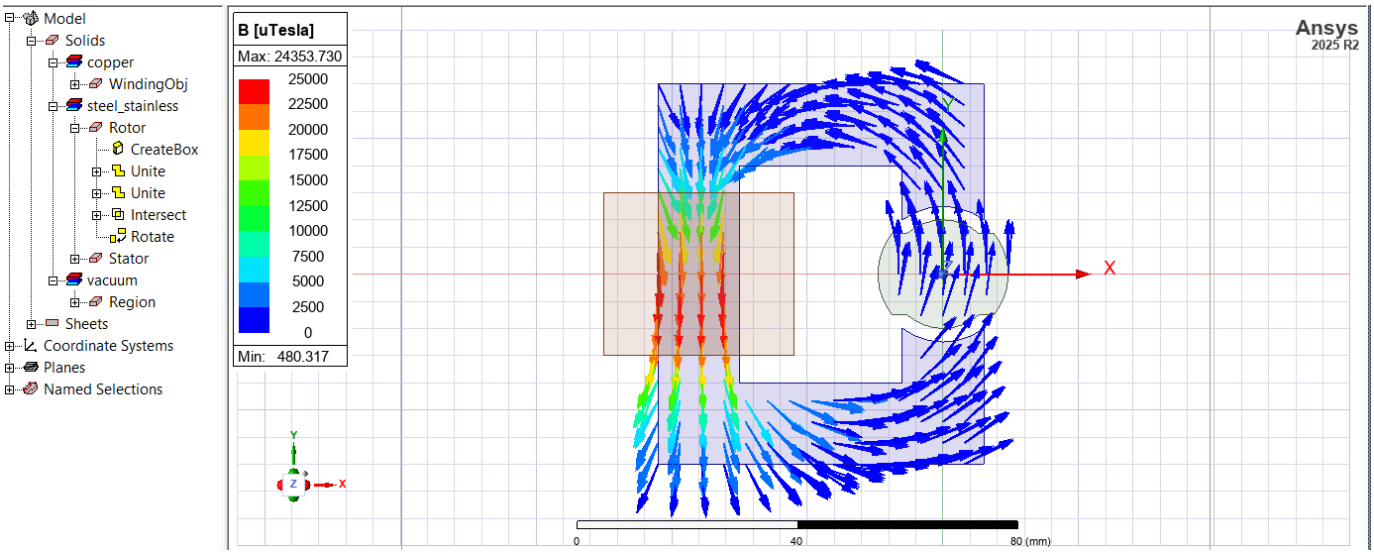


Figure 17 Flux Density Vectors for Rotor Position: 0°

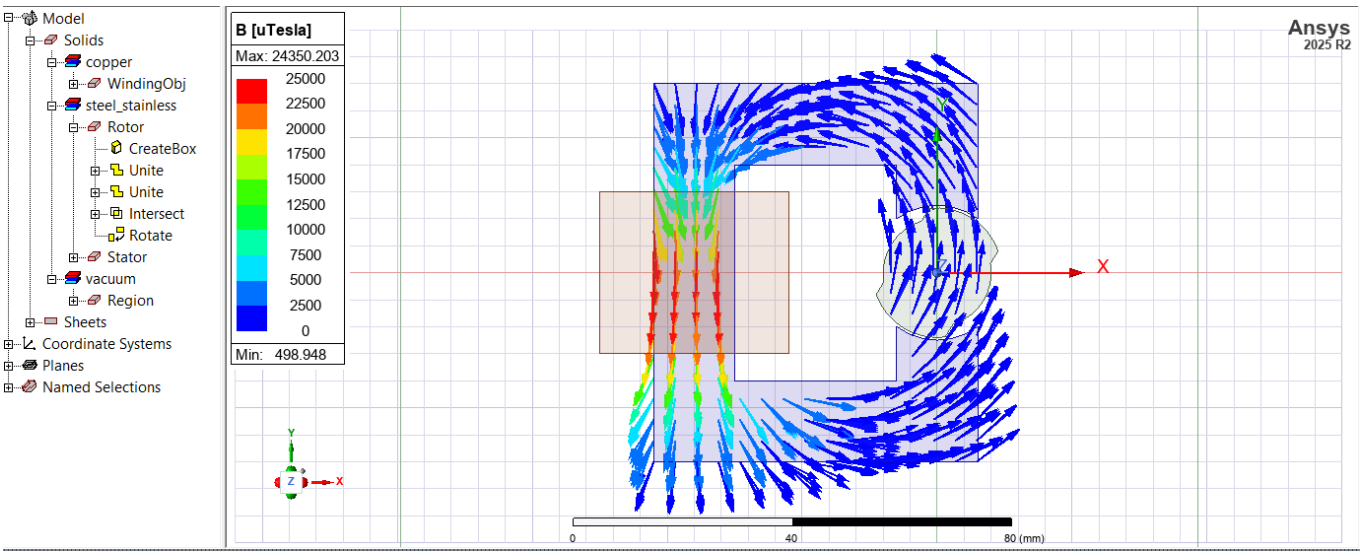


Figure 18 Flux Density Vectors for Rotor Position: 45°

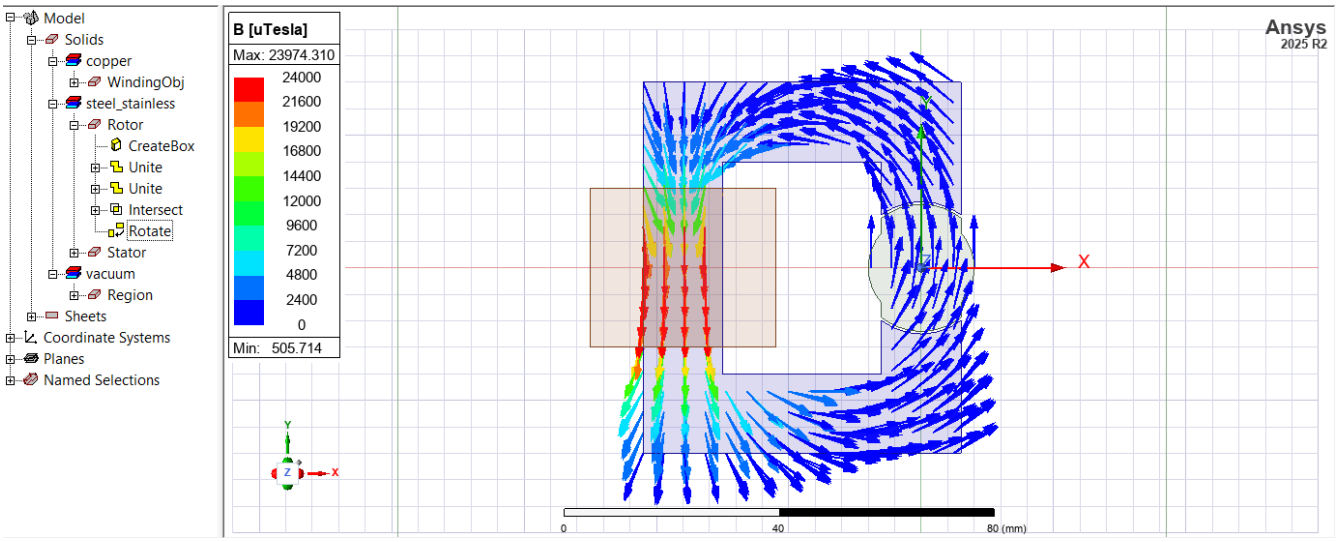


Figure 19 Flux Density Vectors for Rotor Position: 90°

B)

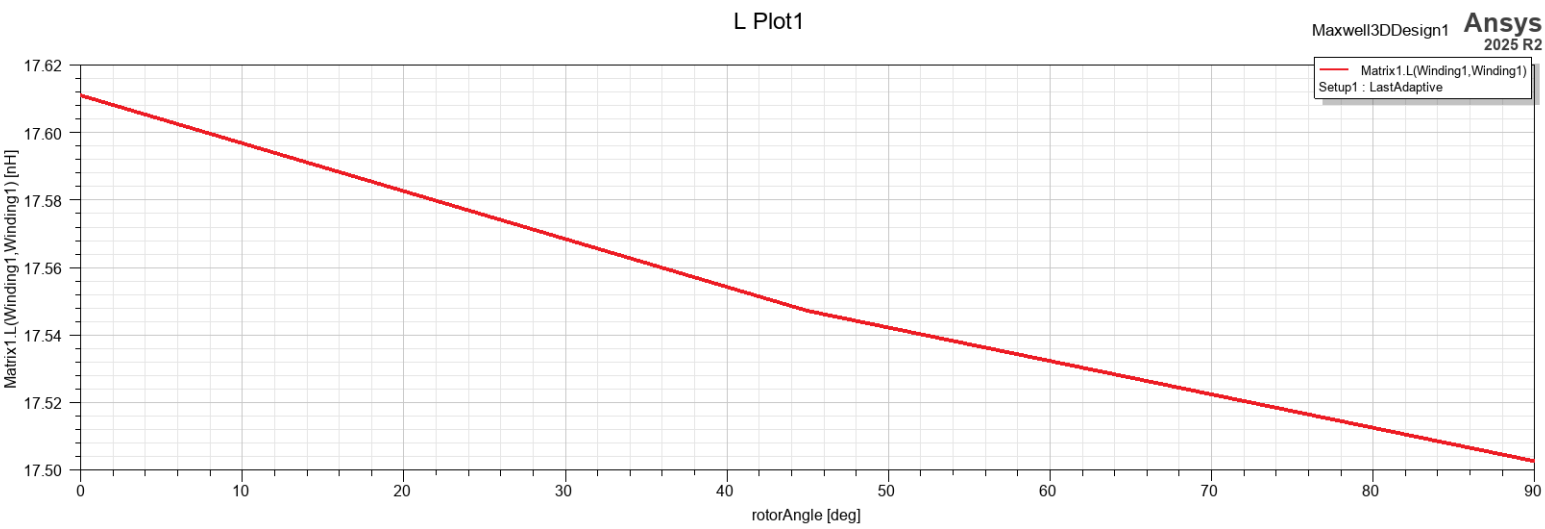
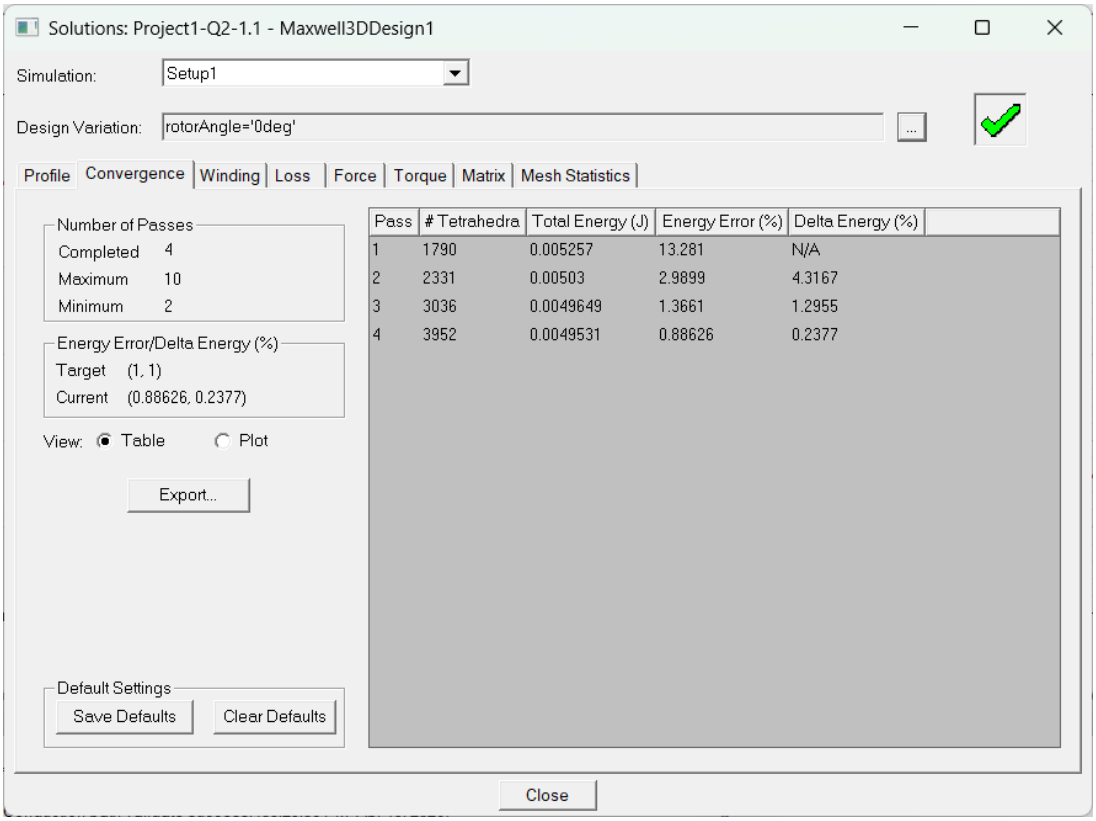


Figure 20 Inductance Values on Rotor Angle 0°, 45° and 90°

The inductance graph shows different characteristics compared to analytical solutions. Since we ignore the reluctance of rotor and stator and material properties in analytical solutions we see that kind of difference. Also note that for real inductance values we have to multiply the values with N^2 (300^2) since we approximate the coils as a single solid cooper object rather than 300 induvial copper cables.



Pass	# Tetrahedra	Total Energy (J)	Energy Error (%)	Delta Energy (%)
1	1790	0.005257	13.281	N/A
2	2331	0.00503	2.9899	4.3167
3	3036	0.0049649	1.3661	1.2955
4	3952	0.0049531	0.88626	0.2377

Figure 21 Stored Energy in The System at Rotor Angle 0°

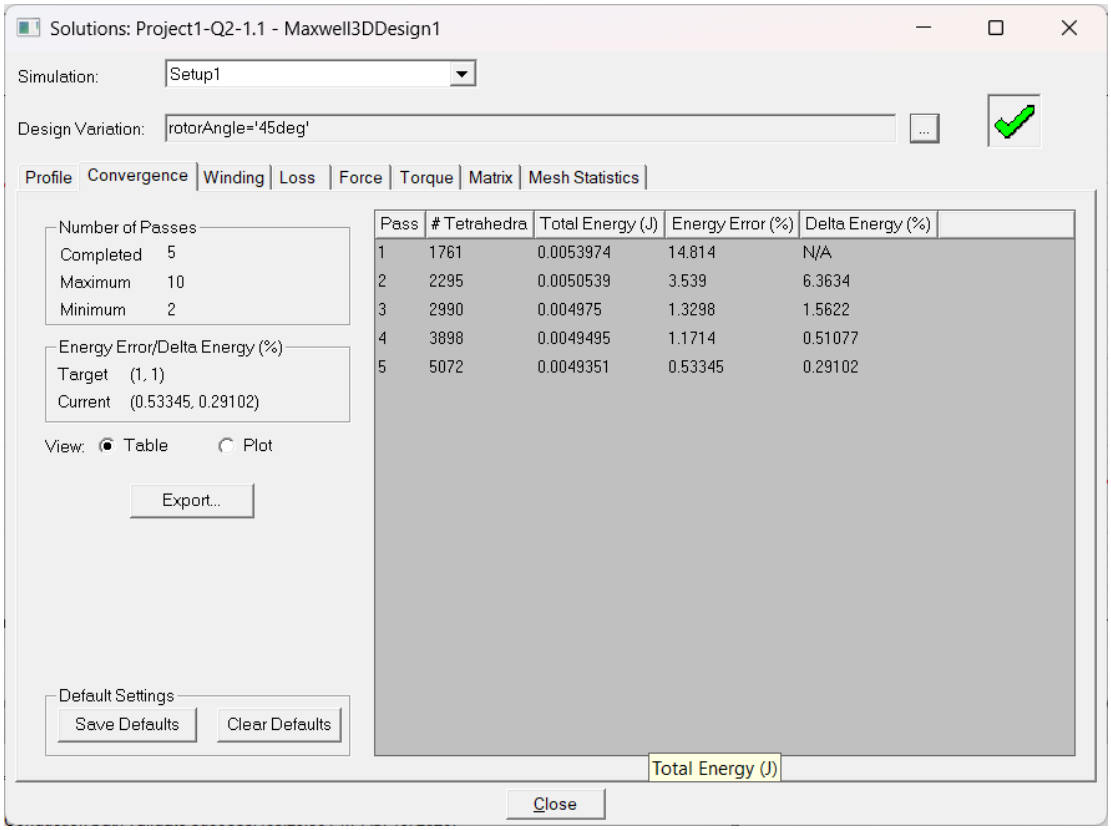


Figure 22 Stored Energy in The System at Rotor Angle 45°

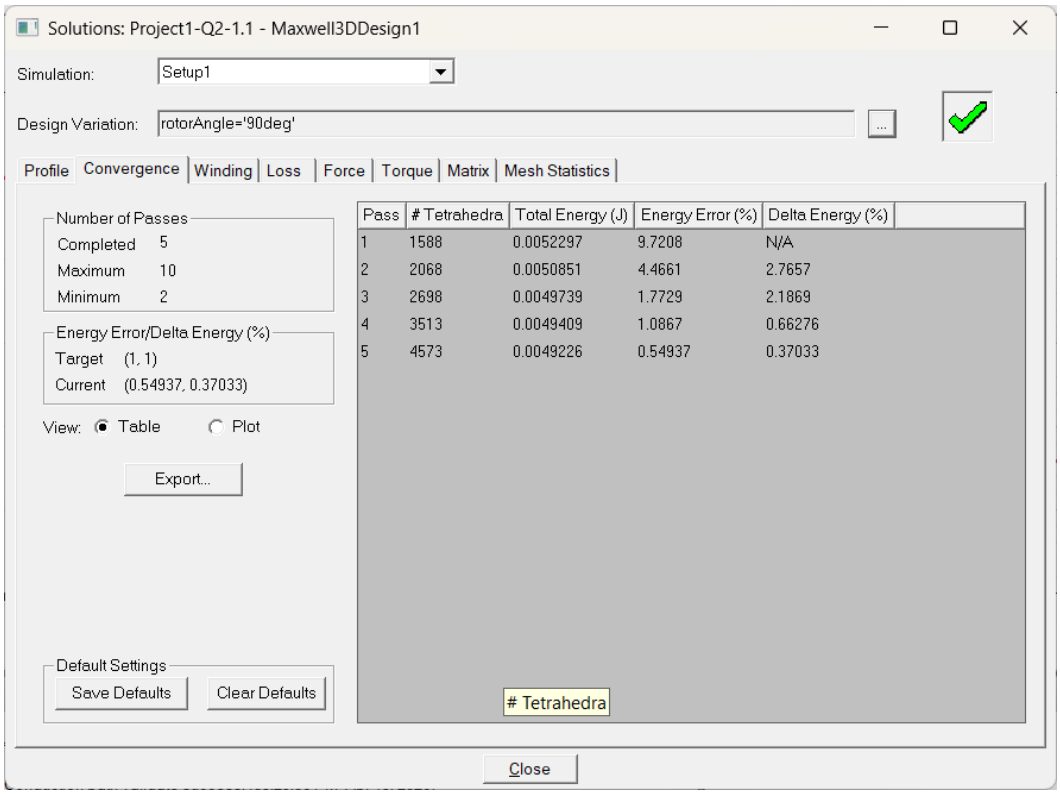


Figure 23 Stored Energy in The System at Rotor Angle 90°

By looking at the total energy stored in the system values, we can say that it follows the same pattern as inductance values we obtain from simulation. The system hold most power when the reluctance values are maximum. At rotor angle 0°, we can see the maximum energy in the system since the maximum reluctance value.

C)

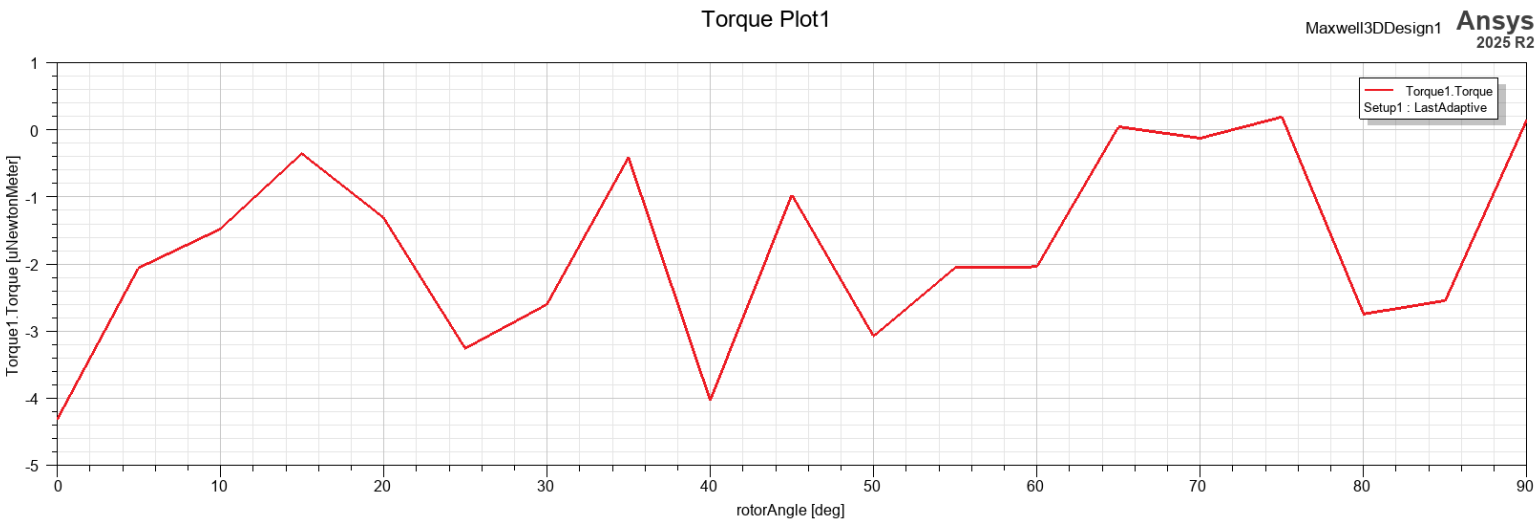
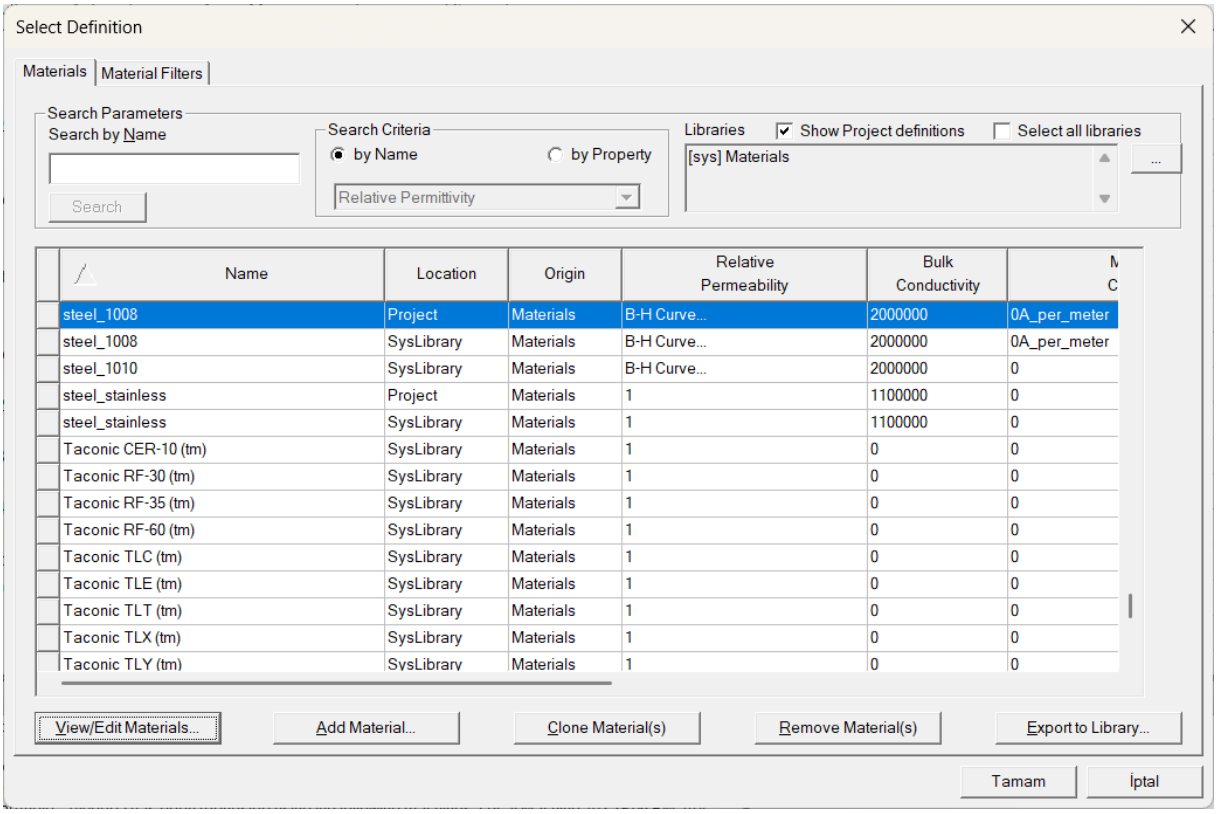


Figure 24 Torque Graph

In torque graph we see saturation which can be caused by Ansys simulation parameters or nonlinear material parameters. Torque results do not match with analytical solutions in Q1.

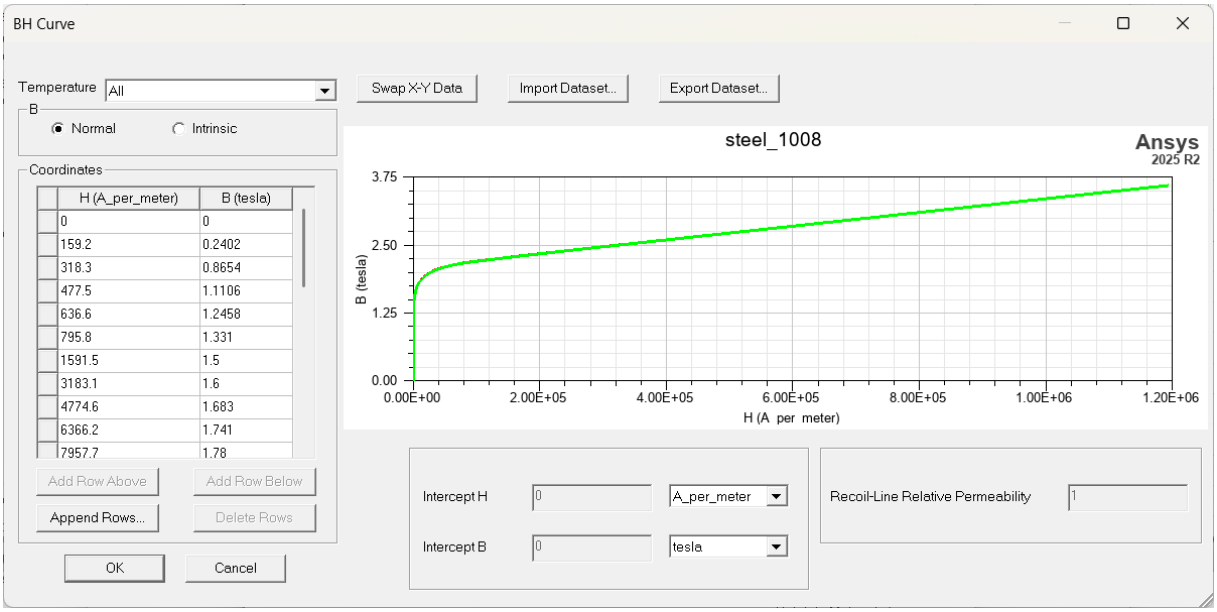
Q3)

In Q3 I do the same simulations as the Q2 but one difference. In Q3 materials are non-linear.



Name	Location	Origin	Relative Permeability	Bulk Conductivity	M C
steel_1008	Project	Materials	B-H Curve...	2000000	0A_per_meter
steel_1008	SysLibrary	Materials	B-H Curve...	2000000	0A_per_meter
steel_1010	SysLibrary	Materials	B-H Curve...	2000000	0
steel_stainless	Project	Materials	1	1100000	0
steel_stainless	SysLibrary	Materials	1	1100000	0
Taconic CER-10 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-30 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-35 (tm)	SysLibrary	Materials	1	0	0
Taconic RF-60 (tm)	SysLibrary	Materials	1	0	0
Taconic TLC (tm)	SysLibrary	Materials	1	0	0
Taconic TLE (tm)	SysLibrary	Materials	1	0	0
Taconic TLT (tm)	SysLibrary	Materials	1	0	0
Taconic TLX (tm)	SysLibrary	Materials	1	0	0
Taconic TLY (tm)	SysLibrary	Materials	1	0	0

Figure 25 Non-Linear Material Selection



H (A_per_meter)	B (tesla)
0	0
159.2	0.2402
318.3	0.8654
477.5	1.1106
636.6	1.2458
795.8	1.331
1591.5	1.5
3183.1	1.6
4774.6	1.683
6366.2	1.741
7957.7	1.78

Figure 26 B-H Curve of Material

Figure 27 Q3 Model (Only Material Difference)

A)

By looking at the flux density vectors on non-linear material we can clearly see that the maximum values increased dramatically due to the magnetic saturation. Also fringing effects are more visible compared to linear materials due to the range of the magnitudes of flux density vectors. Just like on Q2 we can clearly see that maximum flux density achieved on 90° rotor angle while minimum flux density achieved on 0° rotor angle.

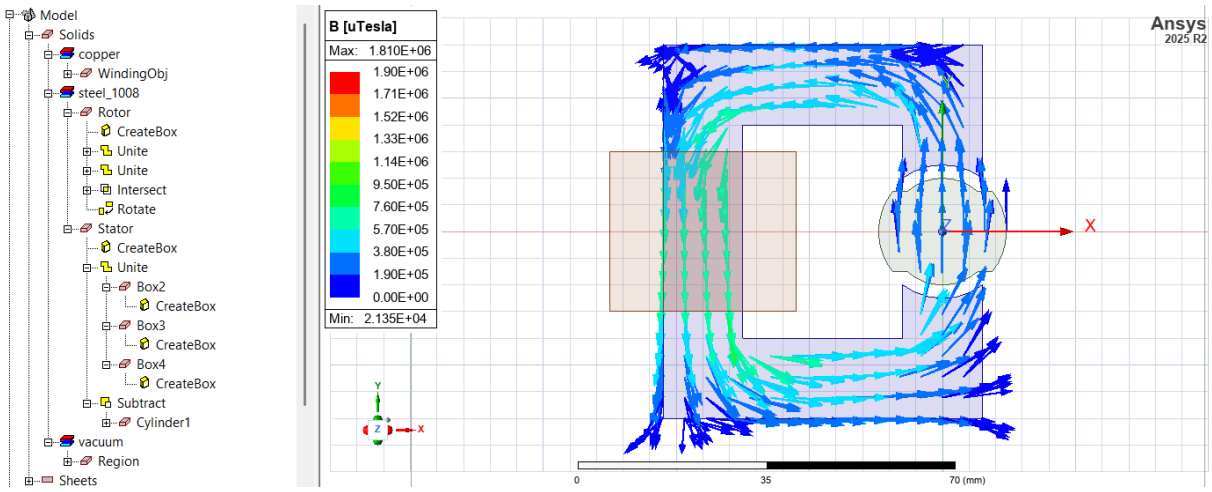


Figure 28 Flux Density Vectors for Rotor Position: 0°

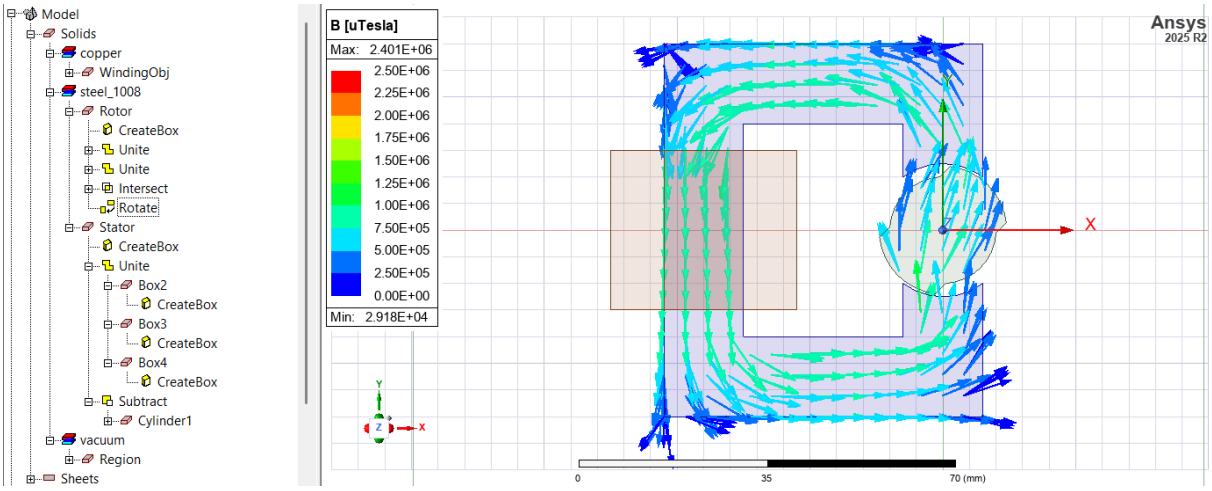


Figure 29 Flux Density Vectors for Rotor Position: 45°

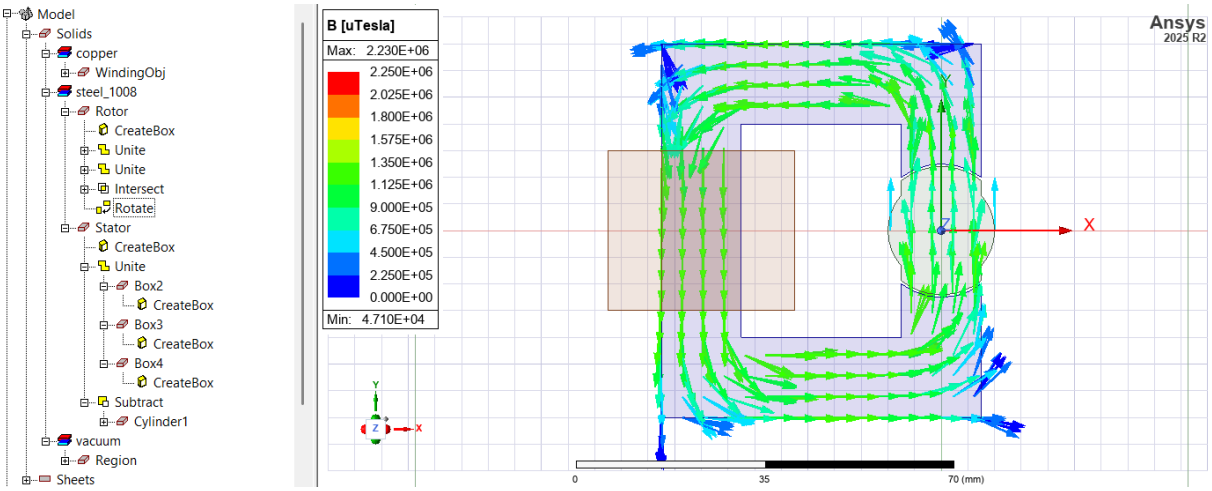


Figure 30 Flux Density Vectors for Rotor Position: 90°

B)

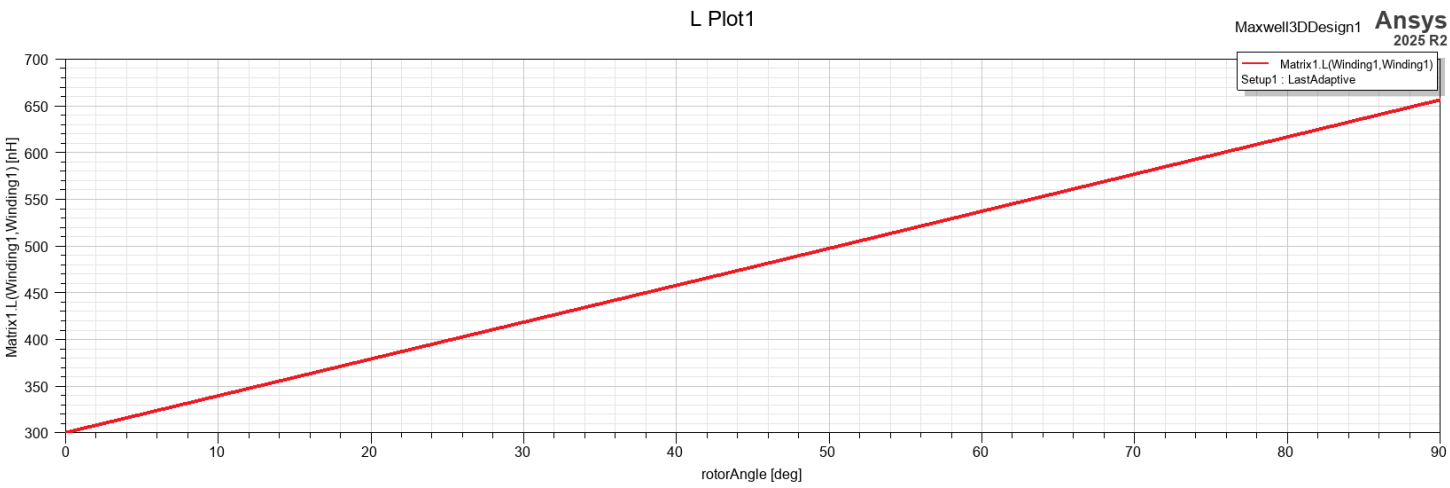
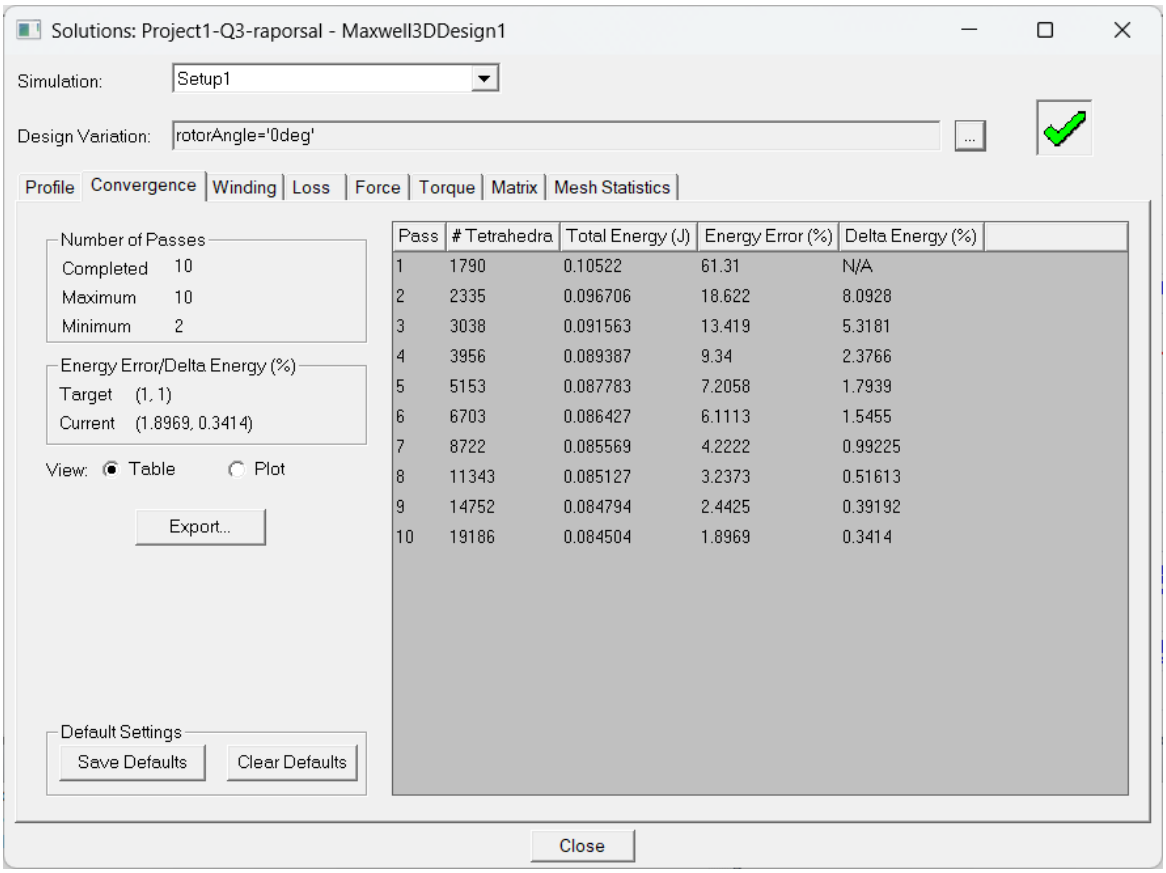


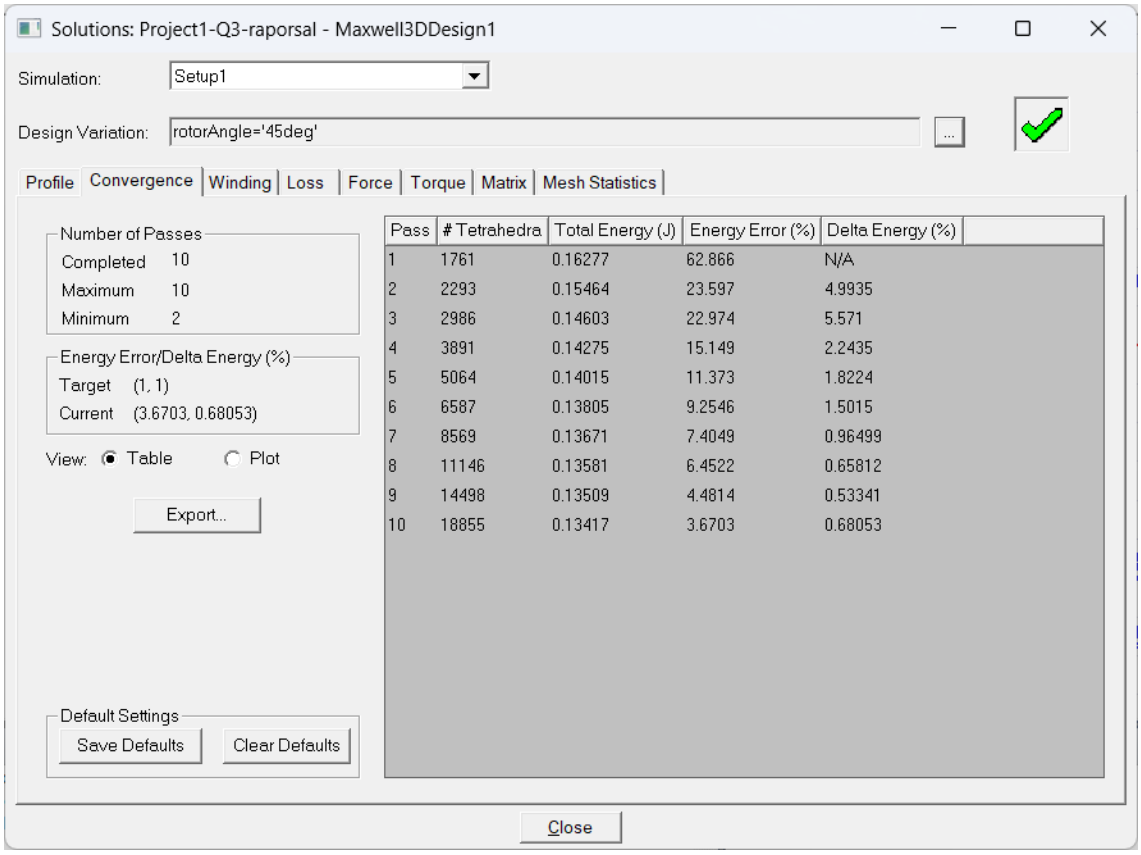
Figure 31 Inductance Values on Rotor Angle 0°, 45° and 90°

The inductance graph shows simular characteristics compared to analytical solutions ın Q1. With the more detailed simulated nonlinear material we can see that higher inductance results less inductance. Also note that for real inductance values we have to multiply the values with N^2 (300^2) since we approximate the coils as a single solid cooper object rather than 300 induvial copper cables.



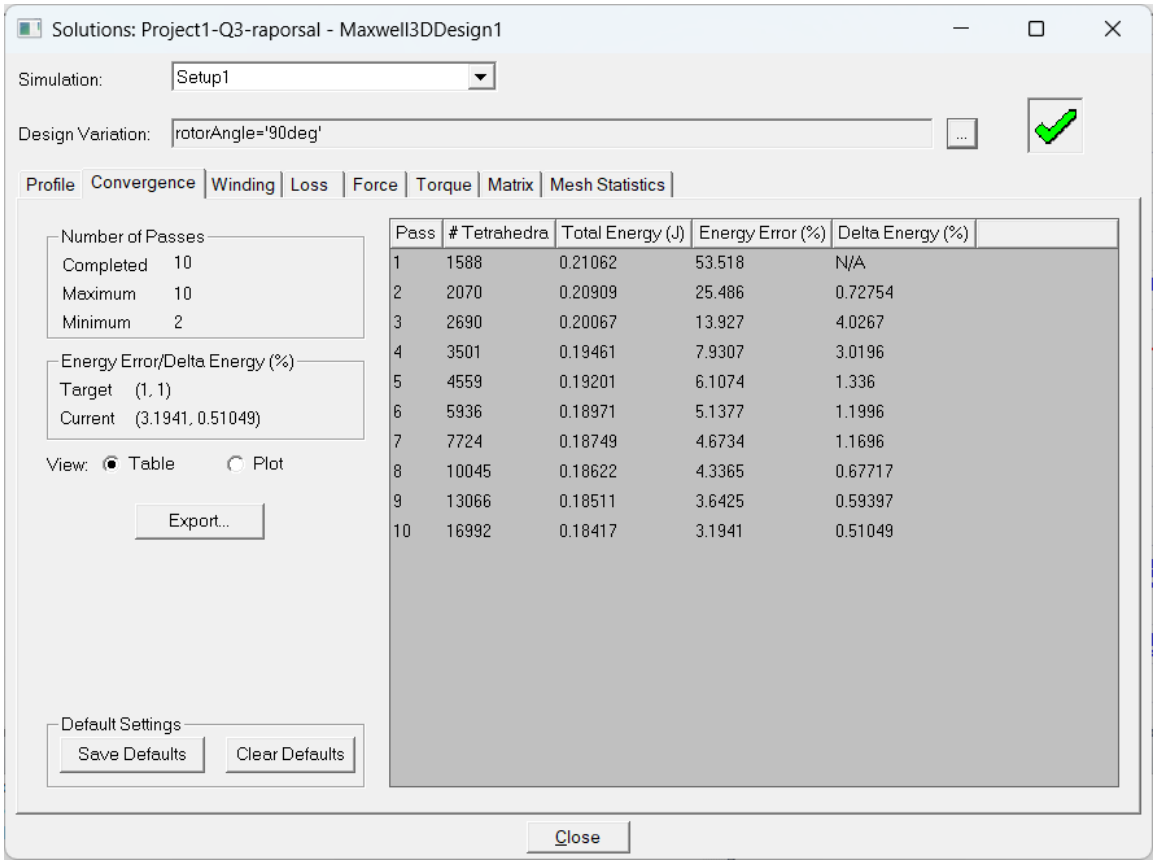
Pass	# Tetrahedra	Total Energy (J)	Energy Error (%)	Delta Energy (%)
1	1790	0.10522	61.31	N/A
2	2335	0.096706	18.622	8.0928
3	3038	0.091563	13.419	5.3181
4	3956	0.089387	9.34	2.3766
5	5153	0.087783	7.2058	1.7939
6	6703	0.086427	6.1113	1.5455
7	8722	0.085569	4.2222	0.99225
8	11343	0.085127	3.2373	0.51613
9	14752	0.084794	2.4425	0.39192
10	19186	0.084504	1.8969	0.3414

Figure 32 Stored Energy in The System at Rotor Angle 0°



Pass	# Tetrahedra	Total Energy (J)	Energy Error (%)	Delta Energy (%)
1	1761	0.16277	62.866	N/A
2	2293	0.15464	23.597	4.9935
3	2986	0.14603	22.974	5.571
4	3891	0.14275	15.149	2.2435
5	5064	0.14015	11.373	1.8224
6	6587	0.13805	9.2546	1.5015
7	8569	0.13671	7.4049	0.96499
8	11146	0.13581	6.4522	0.65812
9	14498	0.13509	4.4814	0.53341
10	18855	0.13417	3.6703	0.68053

Figure 33 Stored Energy in The System at Rotor Angle 45°



Pass	# Tetrahedra	Total Energy (J)	Energy Error (%)	Delta Energy (%)
1	1588	0.21062	53.518	N/A
2	2070	0.20909	25.486	0.72754
3	2690	0.20067	13.927	4.0267
4	3501	0.19461	7.9307	3.0196
5	4559	0.19201	6.1074	1.336
6	5936	0.18971	5.1377	1.1996
7	7724	0.18749	4.6734	1.1696
8	10045	0.18622	4.3365	0.67717
9	13066	0.18511	3.6425	0.59397
10	16992	0.18417	3.1941	0.51049

Figure 34 Stored Energy in The System at Rotor Angle 45°

By looking at the total energy stored in the system values, we can say that the system hold most power when the reluctance values are minimum. At rotor angle 0°, we can see the maximum energy in the system since the minimum reluctance value. Lower reluctance means more dense flux vectors which means more energy.

C)

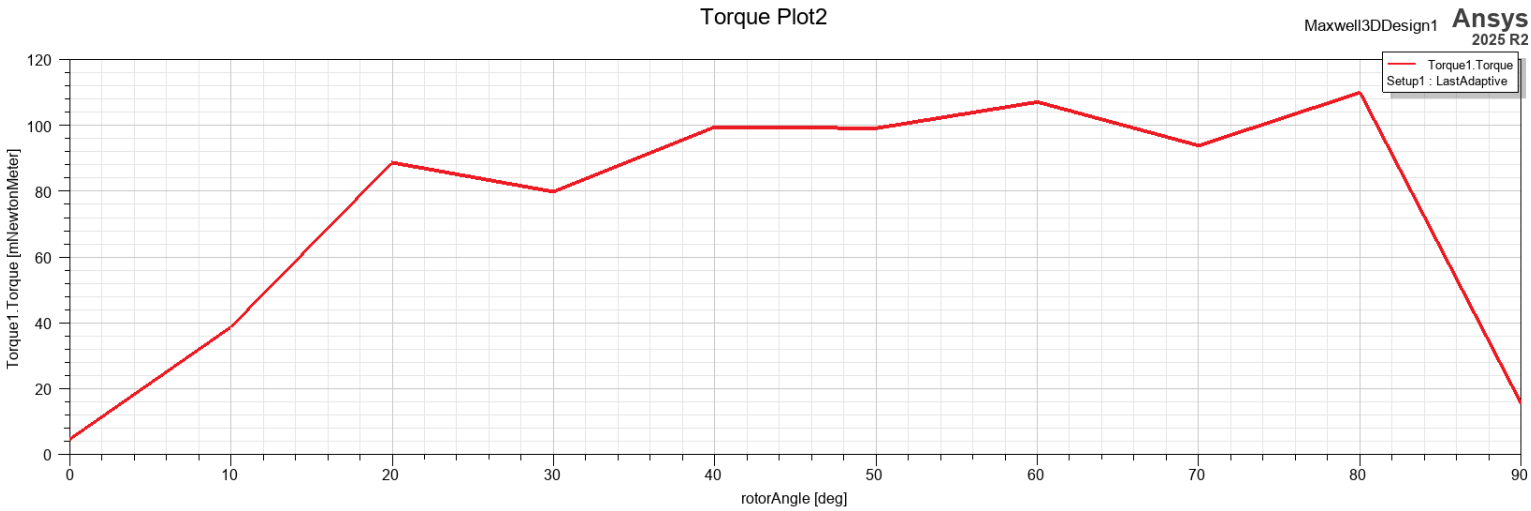


Figure 35 Torque Graph

The torque graph shows us a little rippled sinusoidal. Which is parallel to the analytic part. Compared to Q2 we can see an improvement. As flux density vectors reach higher magnitudes, we can clearly see the improvement on the torque graph.

D)

While the linear model highlights the geometric importance of the reluctance machine model by showing sharp torque transitions driven by fringe fields at the tooth edges the nonlinear model provides a more realistic representation. With saturation the nonlinear model is essential for actual design.

Q4)

Since I explained in Q1 part C, this electric machine is singly-excited variable reluctance machine. The only possible way is applying current with pulses which will apply torque to the rotor when it rotates 90° to 180° and 270° to 360°. In this graph below we can see that situation. Each current pulse is applying torque to the rotor in the direction of rotation.

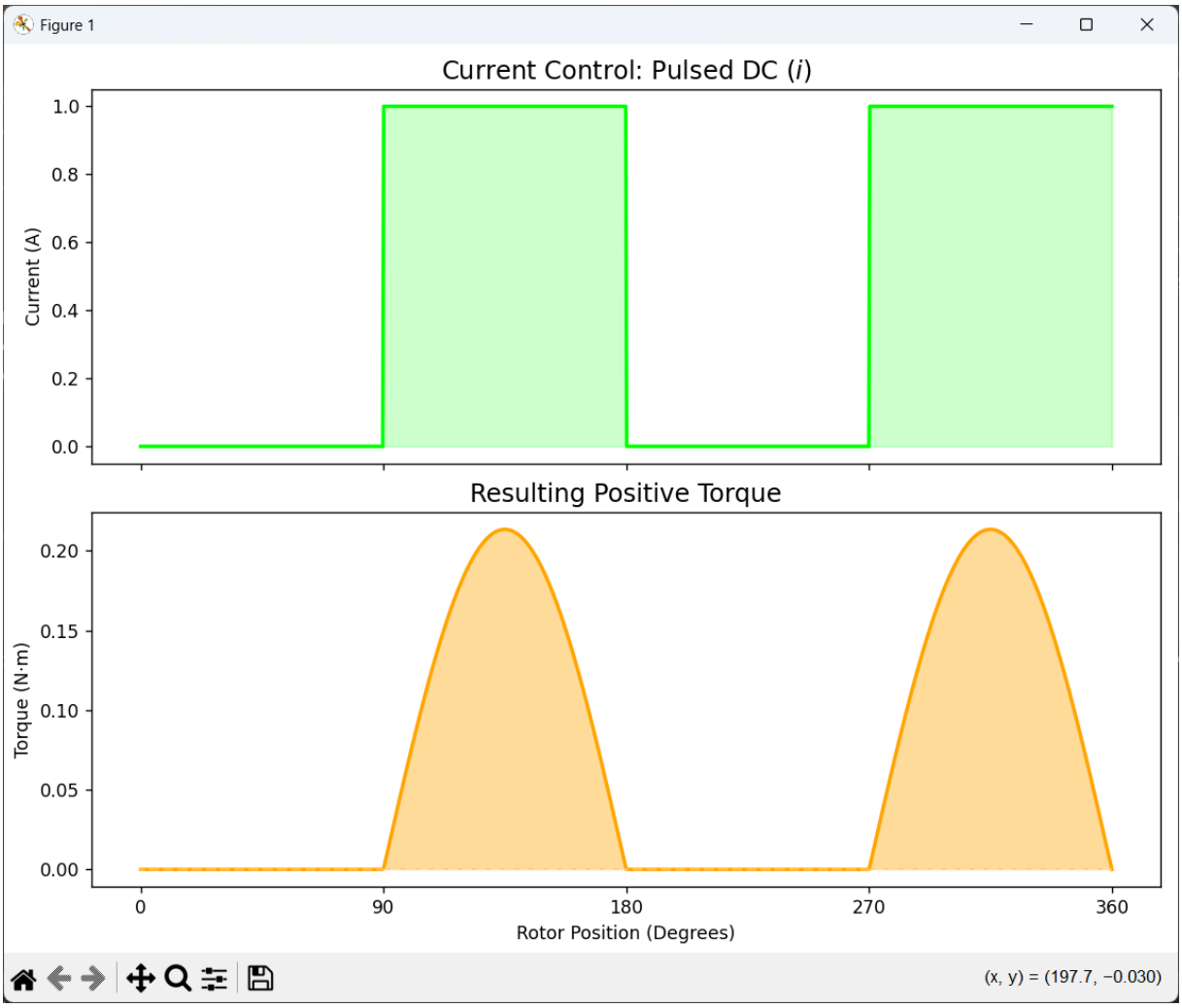


Figure 36 Current Signal for Continuous Rotation of Rotor

Python source code which I used to draw this graph is present on Github repo.

Q5)

To prepare a animation of rotation of rotor, first I adjust the ParametricSetup for rotor angle 0° to 180° with 10° steps. After obtaining the analyze results, using Ansys Maxwell animation function I create the desired animation with desired simulation steps.

Since PDF files does not support .gif files I upload the animated .gif files into Github repo.

Q6)

Since I used Ansys Maxwell for this project all my simulations are 3D from beginning. I explained detailed how I set up the simulation environment and do my simulations.

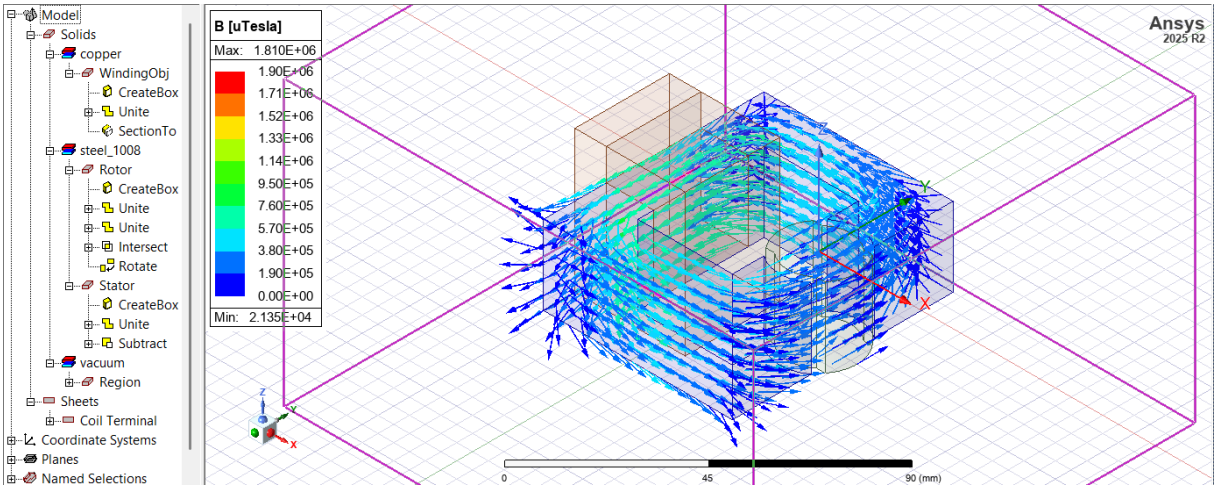


Figure 37 3D Isometric View